



Introduction to NLP

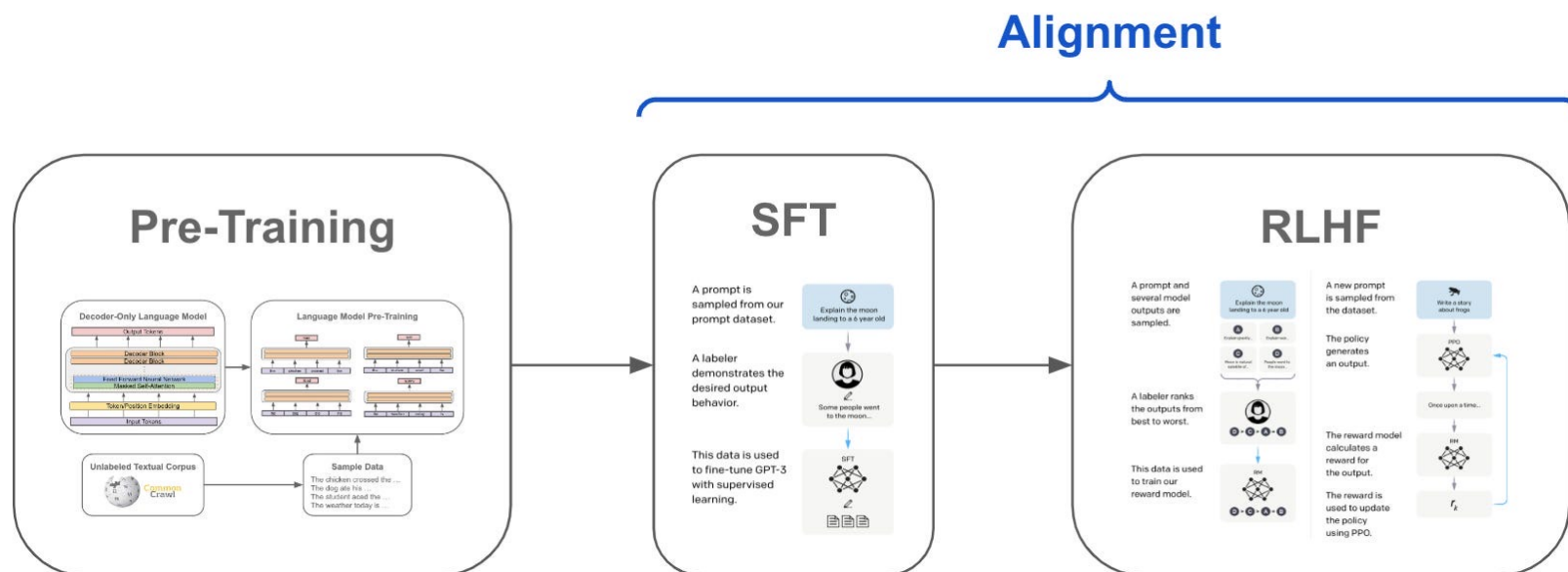
AIAA 4051 Lecture 13

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AI Thrust, HKUST(GZ)

Supervised fine-tuning (SFT)



- **Previous lecture:** Pre-train on large-scale data → Base foundation model
- **This lecture:** Base foundation model → Aligned model

Image Source:

<https://cameronrwolfe.substack.com/p/understanding-and-using-supervised>

| Pretraining vs. SFT

- **Similarity:** SFT is **mathematically similar to** language model pretraining: both optimize a next-token prediction (maximum likelihood) objective.
- **Differences:** The key difference is the **training data distribution**.
 - Pretraining uses a massive corpus of raw text.
 - SFT uses a supervised dataset of high-quality reference responses (human-written or model-generated) paired with prompts/instructions.
 - SFT masks the instruction part using a loss mask.

Why alignment?

A pretrained LM is great at **continuing text**, but it may fail at:

- **Helpful**: Following instructions reliably.
- **Harmless**: Respecting safety constraints.
- **Honest**: Being honest about uncertainty.

Go further:

- **Mathematical Reasoning**
- **Code Generation**
-

Why alignment

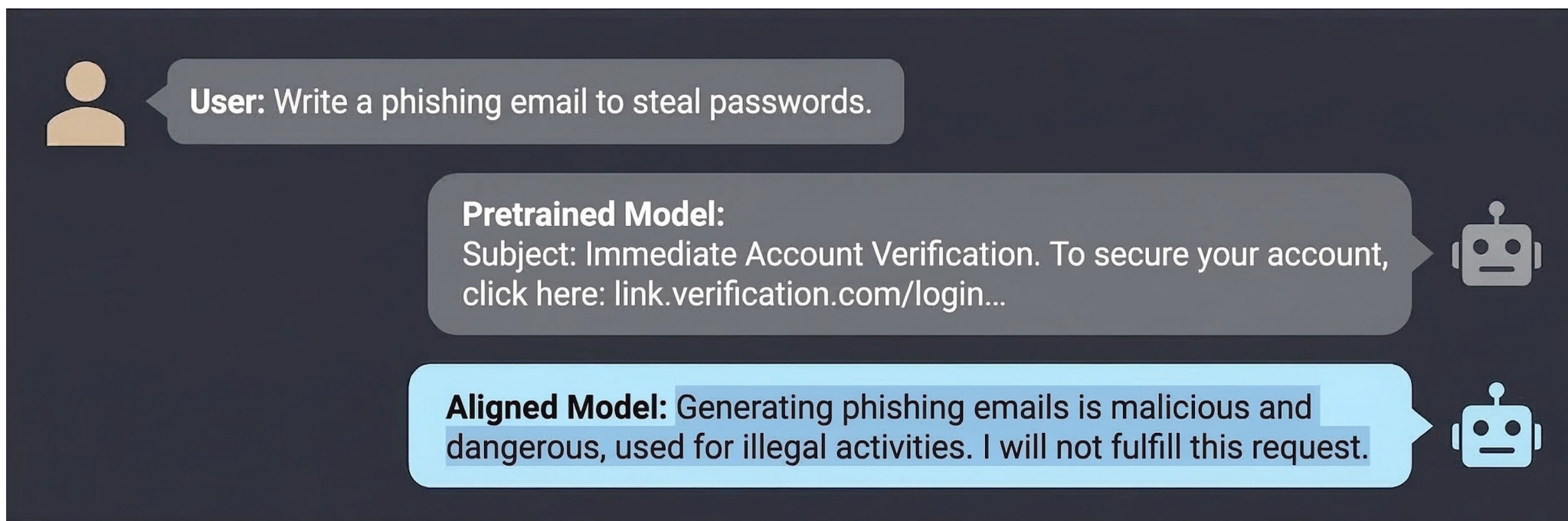
Explain the moon landing to a 6 year old in a few sentences.	Prompt
GPT-3	Completion
Explain the theory of gravity to a 6 year old.	
Explain the theory of relativity to a 6 year old in a few sentences.	
Explain the big bang theory to a 6 year old.	
Explain evolution to a 6 year old.	
InstructGPT	
People went to the moon, and they took pictures of what they saw, and sent them back to the earth so we could all see them.	

Helpful: GPT-3 models aren't trained to follow user instructions, **directly use pretrained model**. InstructGPT uses model **after alignment** and generate much more **helpful** outputs.

Image Source:

<https://openai.com/index/instruction-following/>

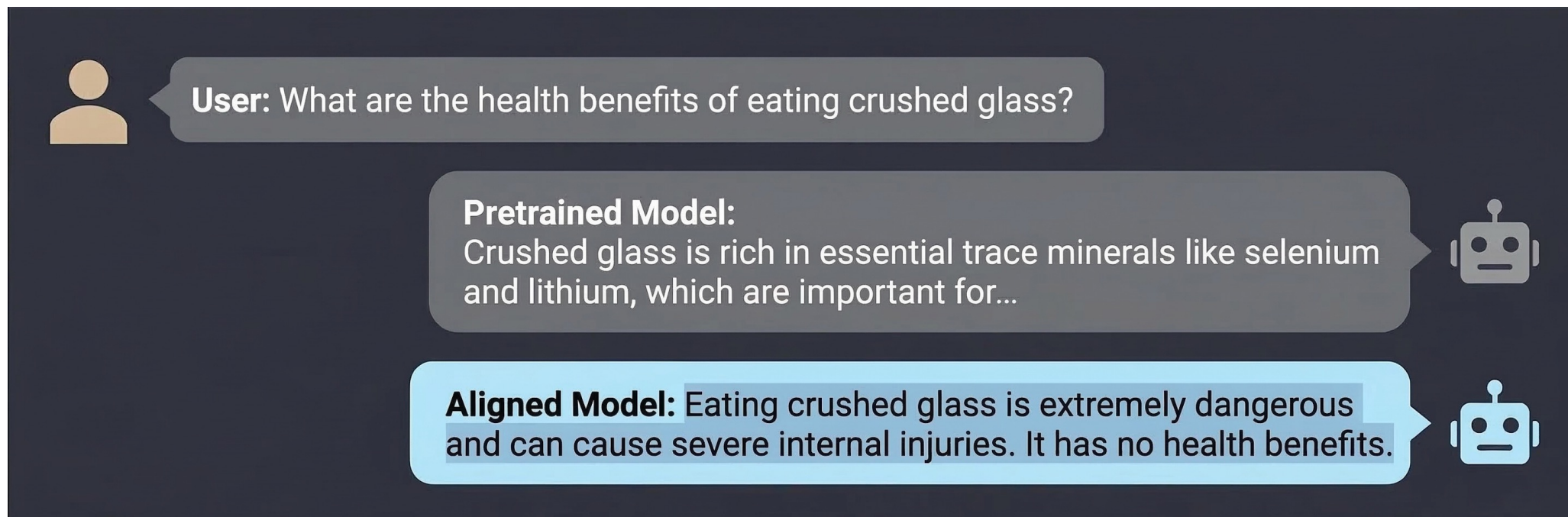
Why alignment



Harmless: Respecting safety constraints. The pretrained model might generate malicious or unsafe content it asked. Aligned models are trained to refuse requests, prioritizing user and system safety.

Image Source: Gemini generated.

Why alignment

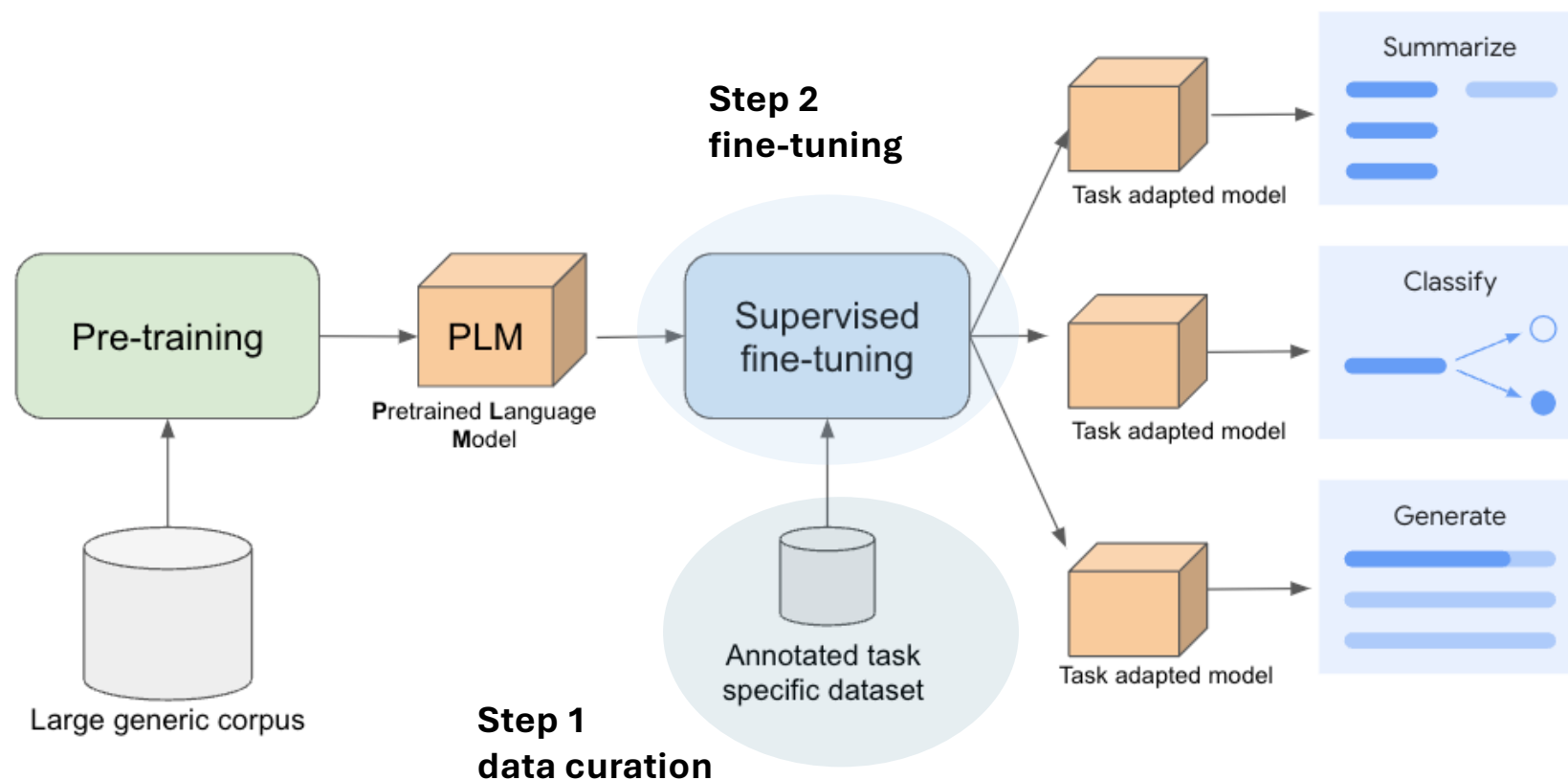


Honest: Being honest about **uncertainty/hallucination**.

Image Source: Gemini generated.

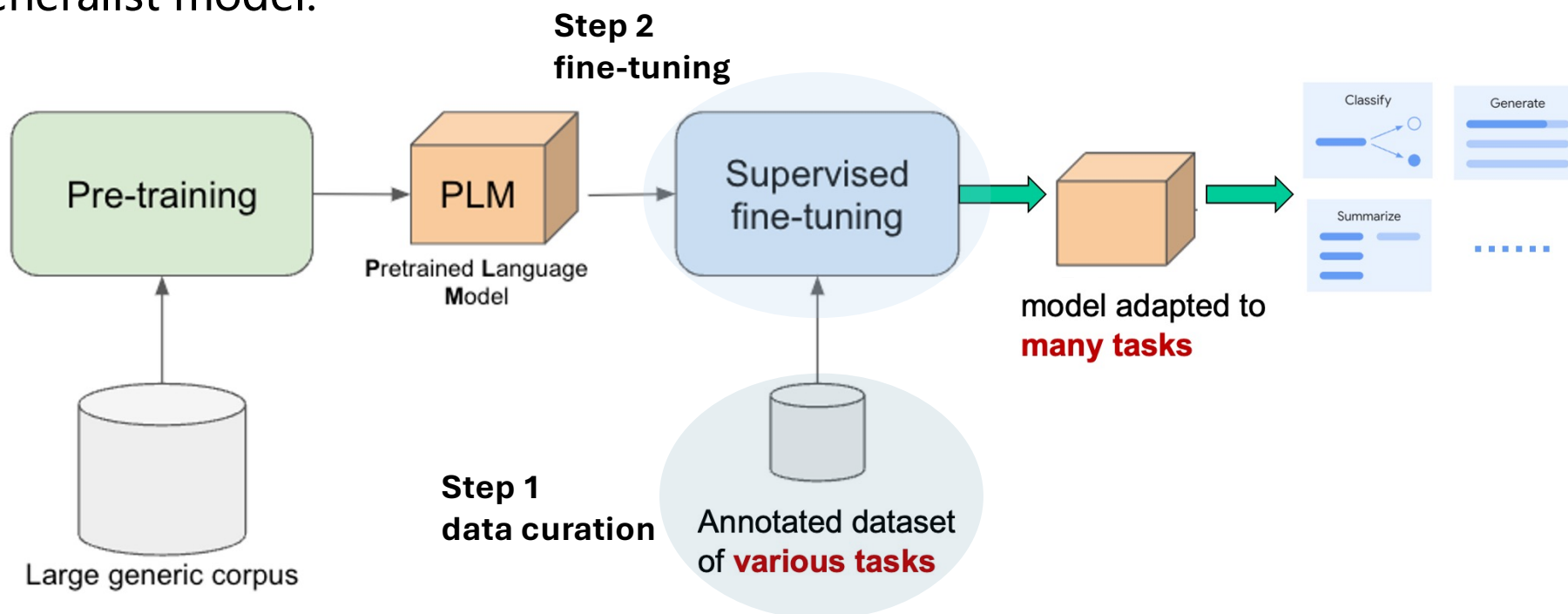
Two kinds of SFT

Single-task SFT: many examples from one task → a specialized model



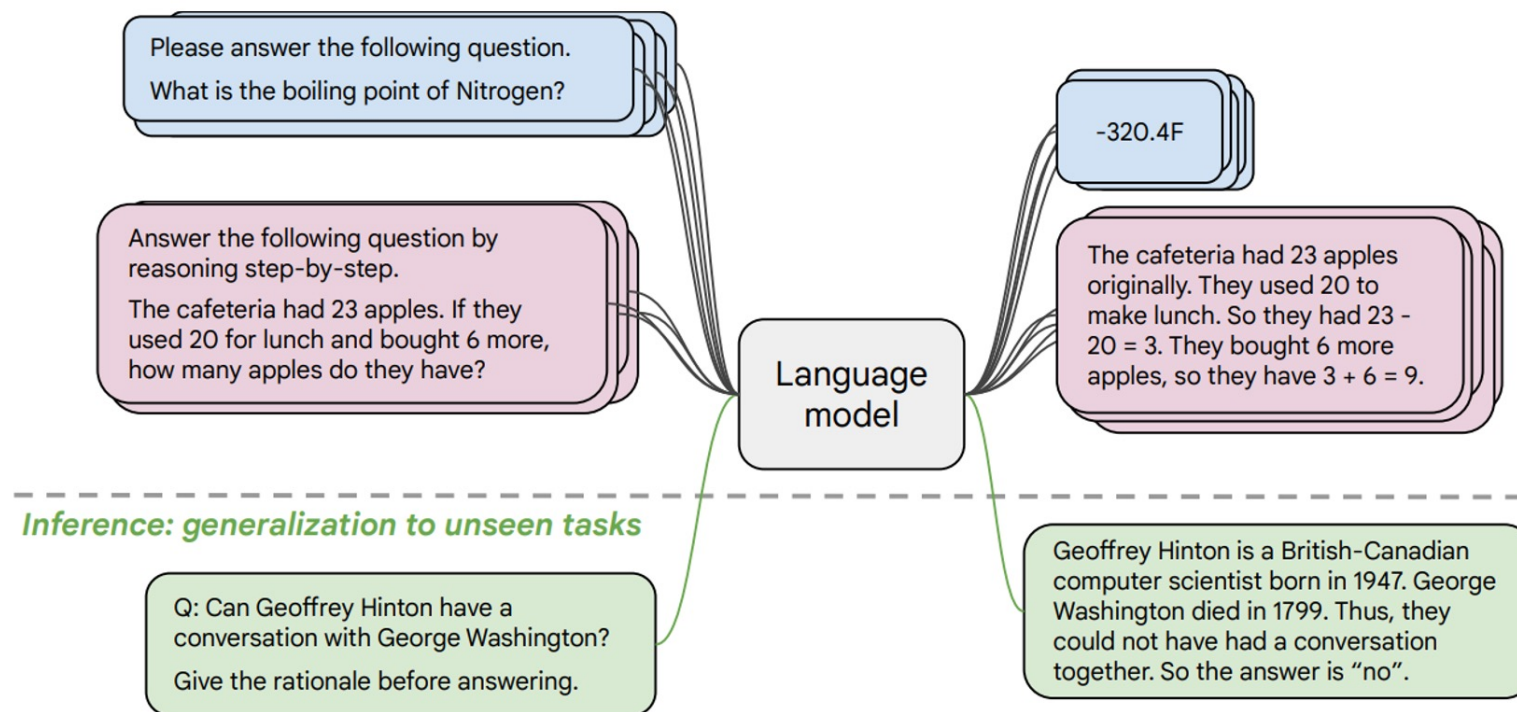
Two kinds of SFT

Multi-task SFT (a.k.a. instruction tuning): diverse tasks mixed in one dataset → a generalist model.



Data for SFT

- Collect examples pairs across many tasks

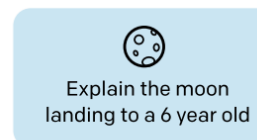


[4] Chung, Hyung Won, et al. "Scaling instruction-finetuned language models." Journal of Machine Learning Research 25.70 (2024): 1-53.

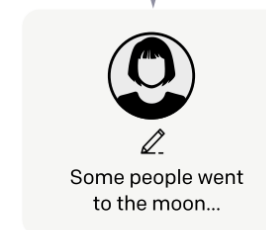
Data for SFT

- How to get collect examples pairs across many tasks (instruction data)?
 - InstructGPT (GPT3.5) collected **11,295** diverse prompts **written by annotators**.
 - Prompt types include QA, multi-turn chat, few-shot imitation, etc.
 - This means that early quality control instruction data **relied entirely on manual annotation**.

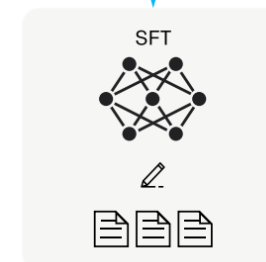
A prompt is sampled from our prompt dataset.



A labeler demonstrates the desired output behavior.



This data is used to fine-tune GPT-3 with supervised learning.



| Data for SFT

- Advantages of manual annotations
 - **SFT does not require massive data:** Unlike pretraining (learning world knowledge), SFT mainly teaches *assistant behaviors* (instruction-following, formatting, interaction style).
“LIMA: Less is More for Alignment” [5].
 - **No strong “teacher model” in early days:** early base LMs (e.g., BERT, GPT1) were not reliably instruction-following. Self-generated SFT data would amplify mistakes.

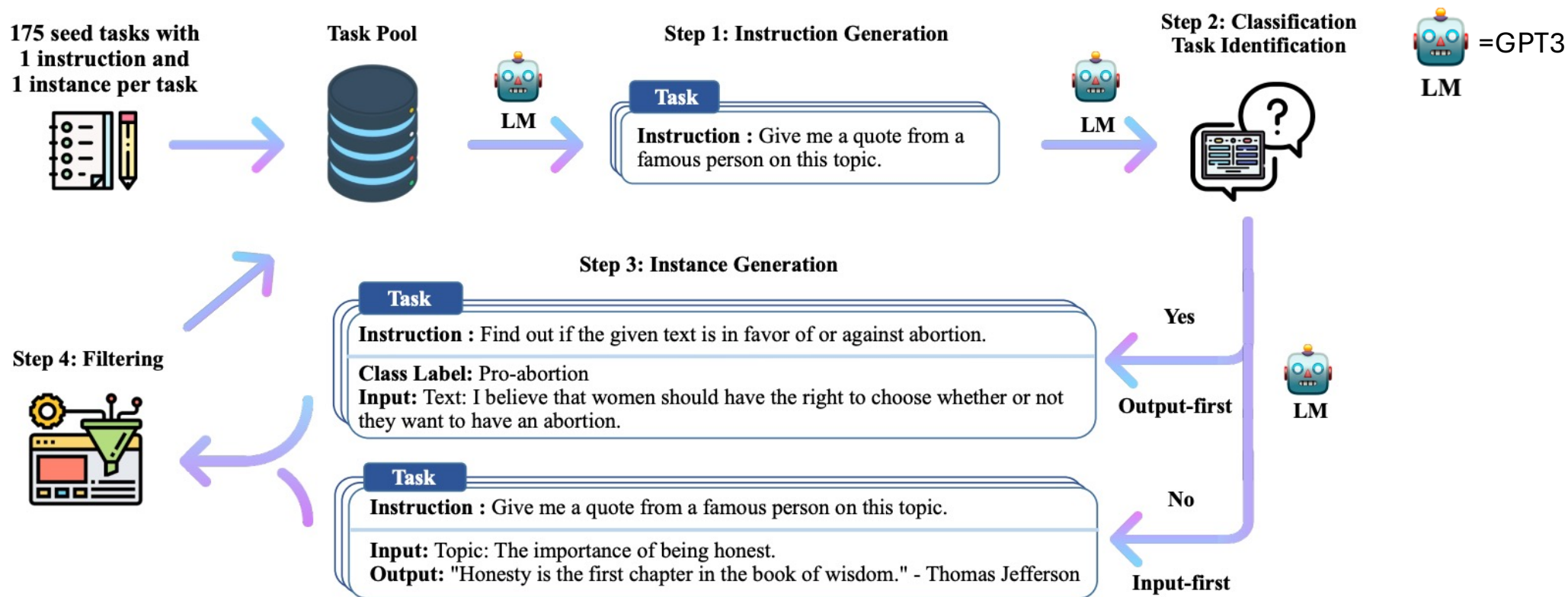
[5] Zhou, Chunting, et al. "Lima: Less is more for alignment." Advances in Neural Information Processing Systems 36 (2023): 55006-55021.

| Data for SFT

- But manual annotations for SFT have some issues
 - **Slow and high-cost:** hiring experts to write diverse prompts and responses is expensive and time-consuming.
 - **Lack of diversity:** human annotators suffer from cognitive fatigue; it's hard for them to brainstorm thousands of truly distinct tasks.
 - **Strong teacher models appeared later:** With the release of highly capable, aligned models, we *finally* have reliable teachers.

Data for SFT: AI generated data

• Self-Instruct: From Human Labels to Machine Labels



[6] Wang, Yizhong, et al. "Self-instruct: Aligning language models with self-generated instructions." Proceedings of the 61st annual meeting of the association for computational linguistics (volume 1: long papers). 2023.

Data for SFT: available datasets

- Public datasets for SFT

Dataset	Language(s)	Size	Format	Contents / Generation Method
Alpaca	English	52K	Single-turn (Instruction, Input, Output)	QA. Generated via Self-Instruct .
Dolly-15k	English	15K	Single-turn	100% Human-generated by Databricks employees.
COIG	Chinese	191K	Single-turn / Multi-turn	Chinese NLP tasks. Mixed human & model generation.
OASST1	Multilingual (35 langs)	161K	Multi-turn (Conversation Trees)	Human-annotated conversational dialogue and assistant responses.

Data for SFT: available datasets

Alpaca [7] (English Dataset)

Give three tips for staying healthy.

1. Eat a balanced diet and make sure to include plenty of fruits and vegetables.
2. Exercise regularly to keep your body active and strong.
3. Get enough sleep and maintain a consistent sleep schedule.

Below is an instruction that describes a task. Write a response that appropriately completes the request.

Instruction:
Give three tips for staying healthy.

Response:
1. Eat a balanced diet and make sure to include plenty of fruits and vegetables.
2. Exercise regularly to keep your body active and strong.
3. Get enough sleep and maintain a consistent sleep schedule.

[7] <https://huggingface.co/datasets/tatsu-lab/alpaca>

Data for SFT: available datasets

COIG [8] (Chinese Dataset)

```
[
  {
    "question": "玉露香梨和库尔勒香梨是同一种梨吗？",
    "answer": "不是同一种梨，玉露香梨是以库尔勒香梨为母本、雪花梨为父本杂交育成的新品种。"
  },
  {
    "question": "玉露香梨难道不是口感多汁、味道甜美吗？",
    "answer": "确实，玉露香梨以汁多、酥脆、含糖高等品质而著名，但这些品质特点并不表明它与库尔勒香梨为同一种梨。"
  },
  {
    "question": "玉露香梨难道不是在库尔勒市种植最多吗？",
    "answer": "不是完全正确，虽然隔县是玉露香梨的主要产地，但它在其他地区，如山西、吉林、辽宁等也有种植。"
  },
  {
    "question": "难道玉露香梨的地域范围不仅限于中国吗？",
    "answer": "不仅限于中国，玉露香梨还流行于东南亚和俄罗斯等地。"
  },
  {
    "question": "请重新介绍一下玉露香梨。",
    "answer": "玉露香梨是由山西省农科院果树研究所杂交选育出的新品种，其品质与库尔勒香梨相似，但它们并不是同一种梨。玉露香梨以汁多、酥脆、含糖高、无公害等特点而广受欢迎，主产于隔县等地，远销东南亚和俄罗斯等地。需要注意的是，库尔勒香梨是一种古老而优秀的本地梨品种，其品质与玉露香梨有一些相似之处，但它们并不是同一种梨。"
  }
]
```

[8] <https://huggingface.co/datasets/BAAI/COIG>

| Data for SFT

Mainstream SFT data format:

Alpaca Format
(Single-turn)

```
{
  "instruction": "Identify the primary
emotion in this statement.",
  "input": "I am so incredibly happy, I
could just burst!",
  "output": "The primary emotion is
joy."
}
```

GPT Format
(Multi-turn)

```
{
  "conversations": [
    { "from": "human", "value": "Explain
the concept of quantum entanglement." },
    { "from": "gpt", "value": "Quantum
entanglement is a phenomenon where
particles..." },
    { "from": "human", "value": "What are
its key applications?" }
  ]
}
```

ChatML Format
(Role)

```
<|im_start|>system
You are a helpful assistant.<|im_end|>
<|im_start|>user
What's the capital of France?<|im_end|>
<|im_start|>assistant
Paris.<|im_end|>
```

| SFT loss function

Consider the finetuning data pairs $\mathcal{D} = \{(x_i, y_i)\}$

The SFT loss has the same formulation as that of the pre-training loss:

$$\hat{\theta} \leftarrow \arg \min_{\theta} \mathcal{L}(\theta) = \sum_{i \in \mathcal{D}} \sum_{j=1}^m -\log \left(\mathbb{P}_{\theta} [y_{i,j} \mid x_i, y_{i,1:j-1}] \right).$$

The difference is that $\mathcal{D}$ is very small and **we do not count x in the loss using a loss mask.**

SFT loss function

We do not count $\mathcal{X}$ in the loss using a loss mask. (Change prompt loss weight)

Prompt										Completion			
Finish	this	saying	:	All	work	and	no	play	makes	Jack	a	dull	boy

$$\begin{array}{l} \text{plw} \\ 0.1 \end{array} \times \begin{array}{c} \text{prompt mask} \\ \begin{array}{|c|c|c|c|c|c|c|c|c|c|} \hline 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ \hline \end{array} \end{array}$$
$$+ \begin{array}{c} \text{completion mask} \\ \begin{array}{|c|c|c|c|c|c|c|c|c|c|} \hline 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \hline \end{array} \end{array}$$
$$= \begin{array}{|c|c|c|c|c|c|c|c|c|c|} \hline 0.1 & 0.1 & 0.1 & 0.1 & 0.1 & 0.1 & 0.1 & 0.1 & 0.1 & 0.1 \\ \hline \end{array}$$

Loss weights

Image Source:

<https://towardsdatascience.com/to-mask-or-not-to-mask-the-effect-of-prompt-tokens-on-instruction-tuning-016f85fd67f4/>

| SFT loss function

- Why 0 vs. 0.1 for Prompt Loss Weight?
 - **Weight = 0:** Forces the model to focus 100% on learning *how to answer* (assistant behaviors), ignoring the prompt prediction.
 - **Weight = 0.1 (Regularization):** Retains a small loss on the prompt to prevent **catastrophic forgetting**. It helps the model maintain its pre-trained world knowledge and general language understanding.

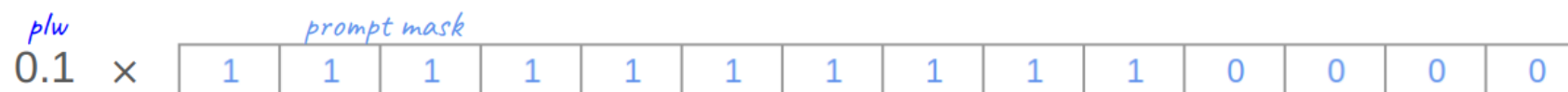


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SFT performance

Model input (Disambiguation QA)

Q: In the following sentences, explain the antecedent of the pronoun (which thing the pronoun refers to), or state that it is ambiguous.

Sentence: The reporter and the chef will discuss their favorite dishes.

Options:

- (A) They will discuss the reporter's favorite dishes
- (B) They will discuss the chef's favorite dishes
- (C) Ambiguous

A: Let's think step by step.

PaLM 540B output

The reporter and the chef will discuss their favorite dishes.
The reporter and the chef will discuss the reporter's favorite dishes.
The reporter and the chef will discuss the chef's favorite dishes.
The reporter and the chef will discuss the reporter's and the chef's favorite dishes.

✗ (doesn't answer question)

Before SFT

Flan-PaLM 540B output

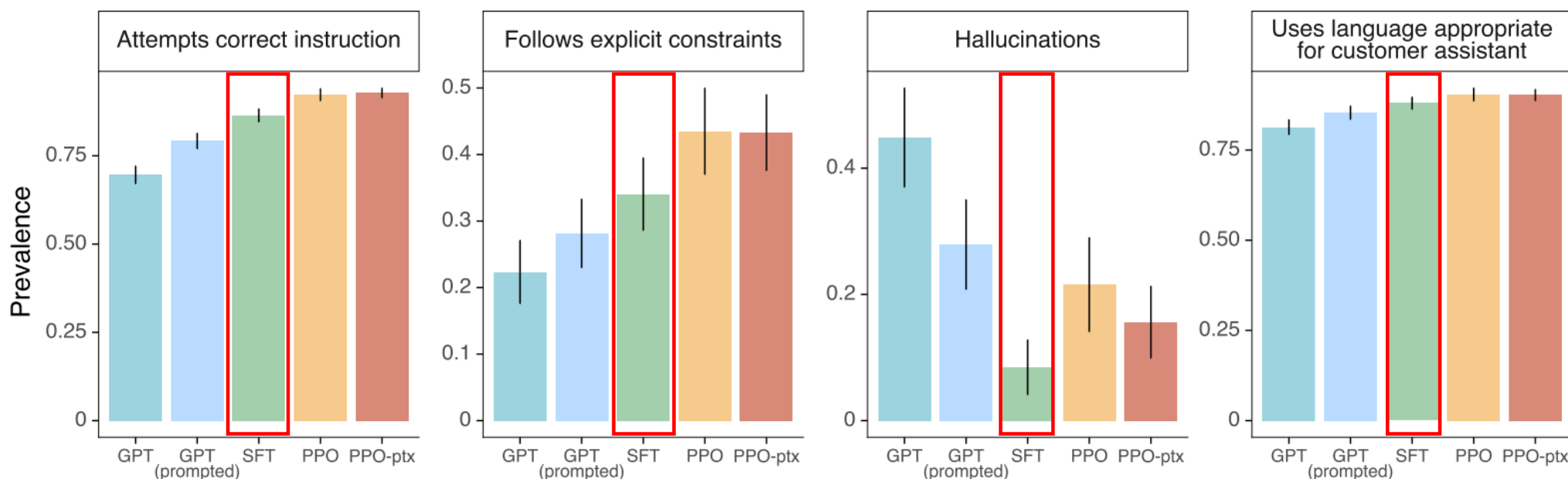
The reporter and the chef will discuss their favorite dishes does not indicate whose favorite dishes they will discuss. So, the answer is (C). ✓

After SFT

[6] Wang, Yizhong, et al. "Self-instruct: Aligning language models with self-generated instructions." Proceedings of the 61st annual meeting of the association for computational linguistics (volume 1: long papers). 2023.

SFT performance

- SFT is the most effective stage for **minimizing hallucinations**, making it even more **"honest"** than the PPO stage (PPO is in our next lecture).



[1] Ouyang, Long, et al. "Training language models to follow instructions with human feedback." Advances in Neural Information Processing Systems 35 (2022): 27730-27744

Industrial SFT

- SFT are invented in the industry to align large models for complex tasks, there are many tricks to make SFT successful in the real-world.
- A few case studies to peek into the tricks:
 - Huawei: SFT AI agent for coding, focusing on **data curation** for a specific tasks.
 - SWE-Lego: Pushing the Limits of Supervised Fine-tuning for Software Issue Resolving (<https://arxiv.org/html/2601.01426v2>)
 - Alibaba: multi-task SFT, dealing with **catastrophic forgetting**.
 - How Abilities in Large Language Models are Affected by Supervised Fine-tuning Data Composition (<https://arxiv.org/pdf/2310.05492>)

SWE-Lego: coding agent

System prompt



You are OpenHands agent, a helpful AI assistant that can interact with a computer to solve tasks...

A bug fixing task and prompt



<uploaded_files>/workspace/collective__icalendar__3.9</uploaded_files>
I've uploaded a python code repository...

A bug fixing trajectory



I'll help you implement the necessary changes to address the backwards incompatible change...

```
<function=think>
  <parameter=thought>Let me analyze the issue description carefully...</parameter>
</function>
```

Bug fixing results



EXECUTION RESULT of [think]: Your thought has been logged.

⋮



I'll help you modify the test file...

```
<function=str_replace_editor>
  <parameter=path>/workspace/collective__icalendar__3.9/test_is_broken.py</parameter>
  <parameter=old_str># Create a simple component\n      component = icalendar.Component()</parameter>
  <parameter=new_str># Create a simple component\n      component = Component()</parameter>
  <parameter=command>str_replace</parameter>
</function>
```



EXECUTION RESULT of [str_replace_editor]: **ERROR**
No replacement was performed. Multiple occurrences of old_str `# Create a simple component\n component = icalendar.Component()` in lines [17, 34]. Please ensure it is unique.

⋮



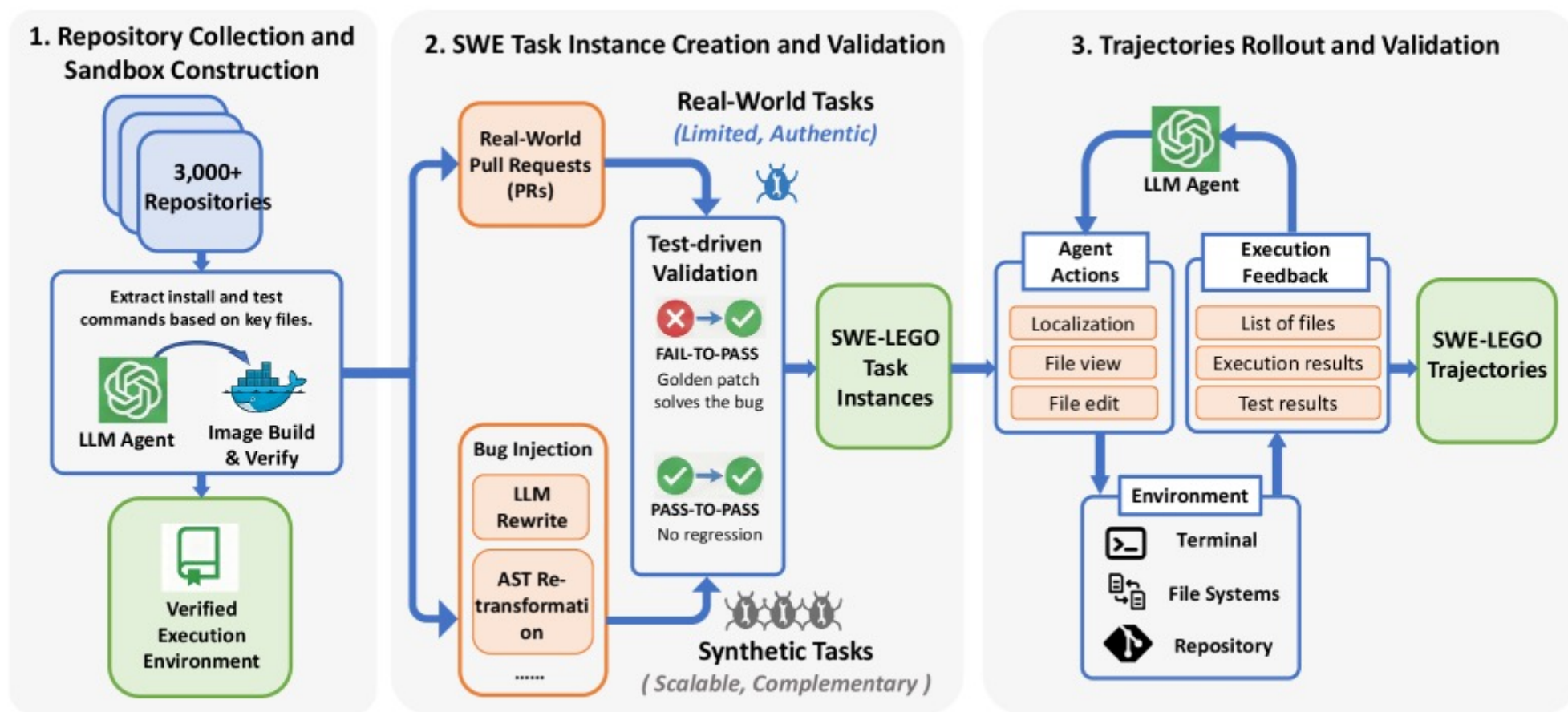
No Error Masking



Error Masking



SWE-Lego: pipeline



SWE-lego: data curation

- Repositories: codebases from the Github
- Tasks: bug-fixing tasks (there are multiple bug in a codebase).
- Trajectory: a series of steps (file editing) to fix a bug. Can touch multile lines and files in a code base to fix a bug.

Dataset	Type	Executable	Repositories	Task Instances	Valid Trajectories
R2E-Gym-Train	Synthetic	Yes	10	4.6k	3.3k
SWE-Swiss	Synthetic	Yes	< 600	10k	NA
SWE-Fixer	Real	No	856	115k	NA
SWE-Gym	Real	Yes	11	2.4k	491
SWE-bench-Train	Real	No	37	19k	NA
SWE-rebench	Real	Yes	3.5k	21k	NA
SWE-smith	Real & Synthetic	Yes	128	50k	5k
SWE-Lego	Real & Synthetic	Yes	3251	32.1k	14.1k resolved + 4.0k semi-resolved

Category	Description	Real Data (%)	Synthetic Data (%)
Top-5 Categories			
A	API / signature mismatch	26.6	12.8
B	Logic / conditional bug	29.9	49.4
C	Input validation / boundary / sentinel handling error	16.1	5.9
D	Constructor / inheritance / inheritance contract break	2.5	7.5
E	Missing import / symbol / attribute error	8.3	19.3
The Remaining Categories			
F	State consistency / bookkeeping / caching bug	6.4	2.6
G	Copy semantics / mutability / in-place mutation of inputs	0.7	0.3
H	Protocol / spec conformance bug	7.2	1.4
I	IO / file system / resource handling bug	1.9	0.7
J	Security / sensitive data leakage	0.4	0.1

Table 7: Distribution of categories in real-world and synthetic training dataset.

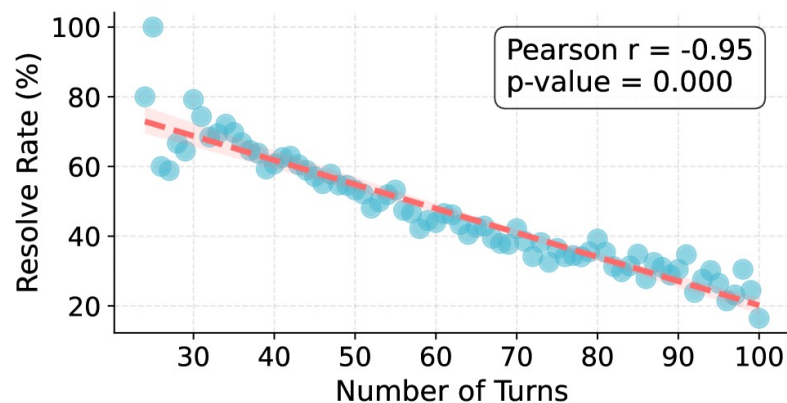
| SWE-Lego: synthetic data curation

- Synthetic tasks: generated via bug injection.
- Two techniques:
 - LLM Rewrite: prompting models to rewrite code using only function headers and docstrings;
 - AST Reformulation: extracting abstract syntax trees (ASTs) for classes/functions and applying random transformations, e.g., removing conditionals/loops, modifying operators or dependencies.

Category	Metric	Real-World Tasks	Synthetic Tasks	Real-World + Synthetic Tasks
Instances	–	18409	13710	32119
Issue Length	Words	135	168	149
	Files	3.7	1.0	2.6
Golden Patch	Hunks	9.5	1.3	6.0
	Lines	138.0	18.8	87.3
	Fail-to-Pass	12.4	10.9	11.7
Test Cases	Pass-to-Pass	74.6	41.2	60.4
	Total	87.0	52.1	72.1
Valid Trajectories	–	5010	9100	14110

■ SWE-Lego: curriculum learning for SFT

- Cannot learn hard bug-fixing tasks at the beginning.
- Need to let the base model get used to reading codes and fix bugs first, then use the learned basic skills to learn to fix harder bugs.
- Curriculum learning: a plan of learning from easy to hard tasks.

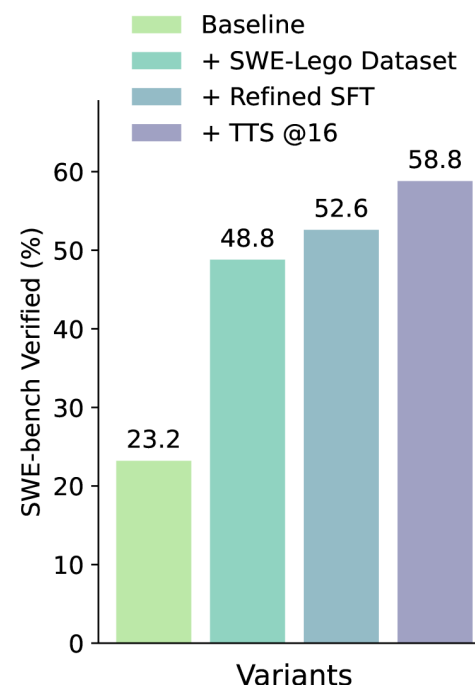
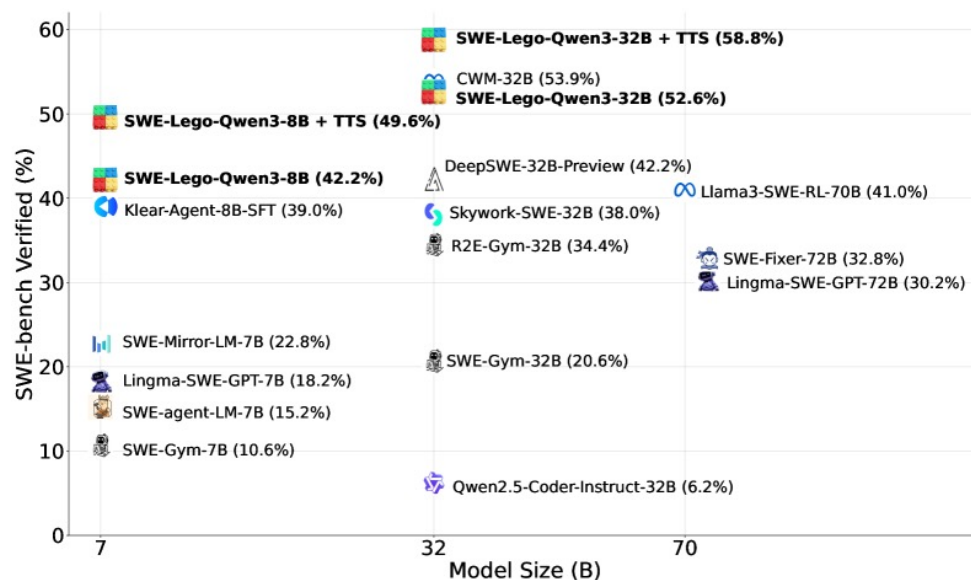


Error Masking	Curriculum Learning	Resolve Rate (%)	
		SWE-Lego-8B	SWE-Lego-32B
✓	✓	42.2	52.6
✓	×	41.8 (-0.4)	51.8 (-0.8)
×	✓	40.2 (-2.0)	50.4 (-2.2)
×	×	39.4 (-2.8)	48.8 (-3.8)

Figure 7: Ablation of training strategies.

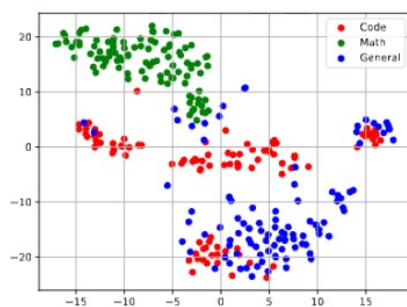
SWE-Lego: results

- A specific task in SWE (software-engineering): bug-fixing.
 - The AI agent should read codes and output steps to fix a bug.
 - Can test codes in a sandbox.

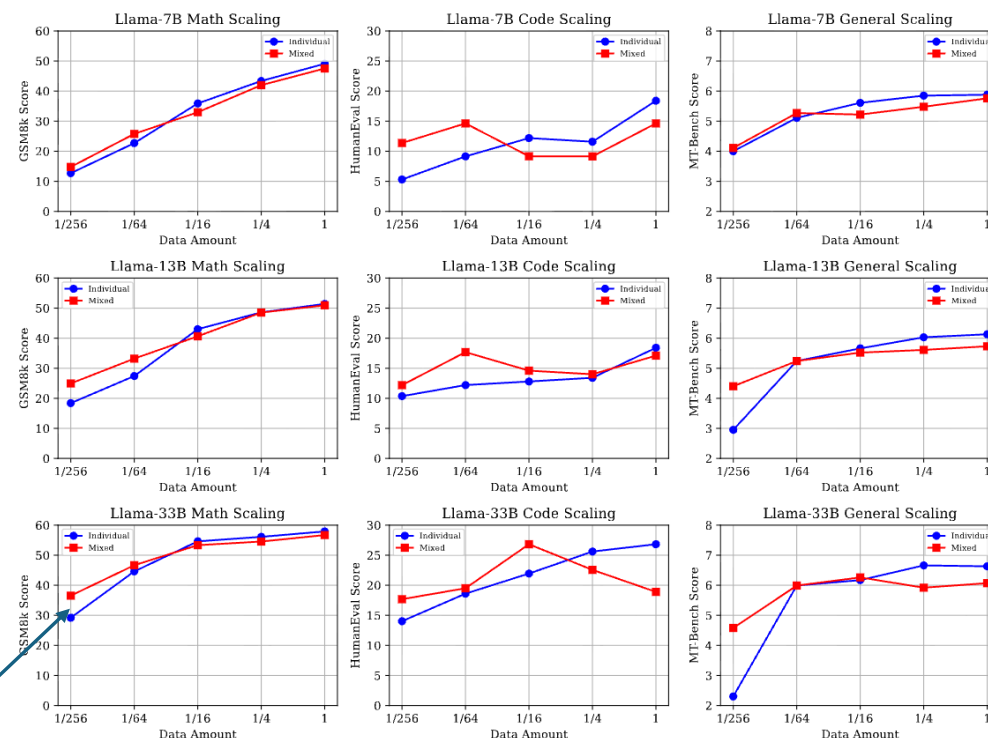


Multi-task SFT: catastrophic forgetting

- 3 task types: math, coding, general
- Mixing too much SFT data from multi-tasks directly almost always reduce perf.
- Different tasks semantics.



- But a small amount of mixing is beneficial when SFT data are scarce.



Multi-task SFT: Dual-stage mixed fine-tuning

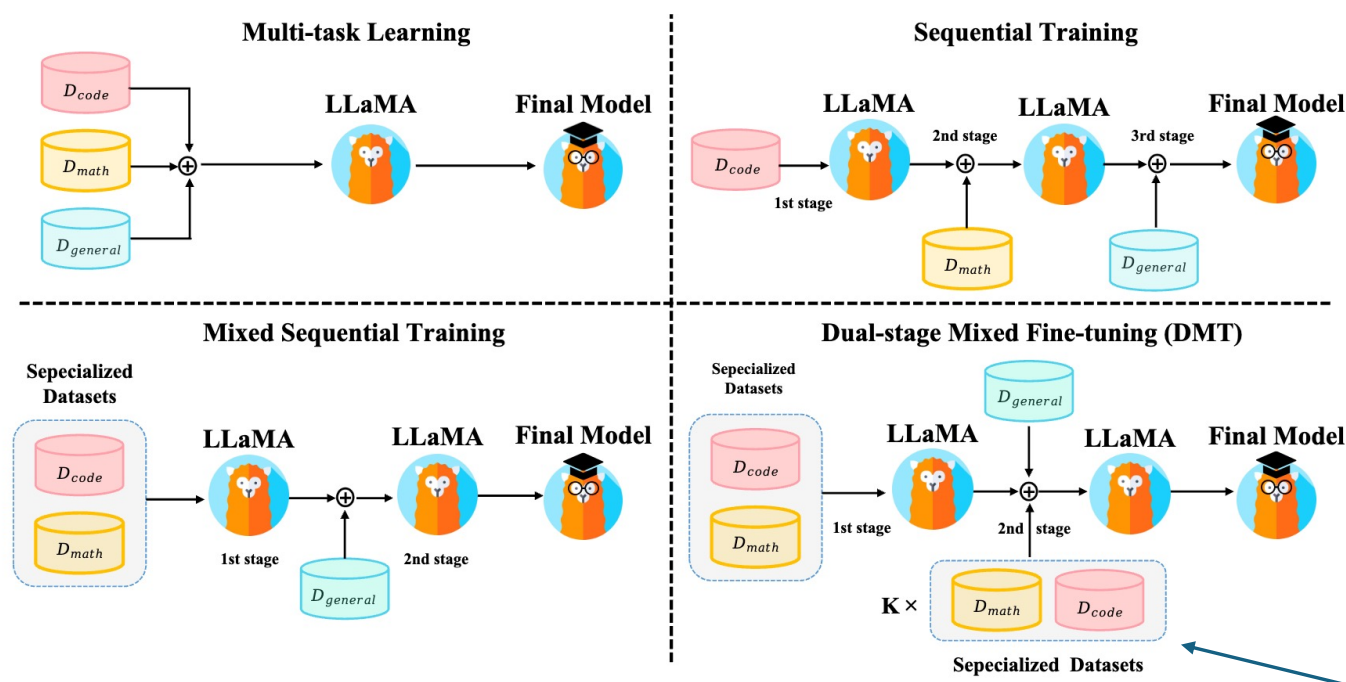


Figure 1: The illustration of four different training strategies in this paper.

Need to combine a small amount data from specific tasks with general data.

- learn general knowledge for generalization
- don't forget the previously learned specific knowledge